C-DISC TECHNOLOGIES · STRATEGY DOCUMENT

The GTM Agent

— how we built it, how it works, and the sources behind every claim.

A live, working go-to-market tool for C-DISC's North-India expansion. Built as the operational layer underneath the strategy deck. Single-page web app, no backend, deployed to Vercel. **This footnoted edition cites the source for every external claim.**

LIVE	cdisc-north-engine.vercel.app [28]
STACK	Vanilla JavaScript · HTML · CSS · OpenAI Responses API [5]
SIZE	~2,500 lines of code · 4 files · ~140 KB
STATE	localStorage [11] (single-user) · no backend database
HOST	Vercel static [9] (free tier, global CDN)
VERSION	v2.0 — footnoted edition · 27 numbered references

Read time: ~14 minutes · Sections: 15 · Pages: ~14 · References: 27

Contents

1. Executive Summary

The **C-DISC North Engine** is a live, working go-to-market (GTM) tool built as the *operational layer* underneath the North-India expansion strategy. It is not a slide deck and not a wireframe — it is a production-deployed single-page web application that a North-India sales team could begin using on day one.

In one sentence: a CRM + lead-discovery + outreach generator + activity tracker, wrapped into one browser app that runs without a backend.

The agent solves a real workflow problem. The moment C-DISC[1] decides to expand into North India, the field team needs to find leads, decide which ones are worth pursuing, generate personalised outreach, track every interaction, and measure pipeline progress. Most of those needs are typically met by a stack of expensive SaaS tools — HubSpot, Lemlist, Apollo, Outreach. The agent collapses them into one focused, single-purpose application tailored to C-DISC's specific constraints (PEN Foundation's segment fit[2], G+1 enforcement, Hindi/English bilingual outreach, North India zones).

2. Why a Tool — Not Just a Strategy Document

A strategy deck tells the C-DISC leadership *what to do*. A working tool tells the **operations associate** *how to do it every morning*. The two are complements, not substitutes.

- A strategy deck without an operating layer dies in a slide reel. The first ops hire still has to figure out their daily workflow from scratch.
- Most North-India construction leads are not in any single database — they need to be discovered (web search), enriched manually, qualified, and then approached in their language.
- C-DISC's product[2][3] has a very specific constraint (the G+1 rule, the segment fit) that no generic CRM enforces. Building a custom tool *embeds* that constraint into the workflow itself.
- A live web app is also a **strategic signal** to leadership: this isn't just a recommendation, the execution layer already exists.

3. What It Does — The User Workflow

```
Discover   →  Discovery Pool  →  CRM  →  Outreach  →  Activity log
(real OpenAI    (user reviews     (selected leads     (queue + send +
 search,         and picks)        with stage track,    mark-sent +
 manual entry,                     drawer, history)     export)
 sample batch)
```

The critical design choice — and the one that separates this from naive "agent" tools — is the **discovery pool**. Discovered leads do **not** automatically enter the CRM. The user has to explicitly check the boxes for the leads they want and click *Add to CRM*. This mirrors how real sales teams work: discovery is noisy, the CRM is the source of truth, and human judgment sits between the two.

4. The Five Views Explained

4.1 CRM (Pipeline Overview)

Five KPI cards (total leads, new, contacted, replied, won), a toolbar with Kanban/Table toggle plus filters, a seven-column kanban board for the pipeline stages, a stage funnel chart, and a recent-activity feed. Clicking any lead opens a **side drawer** with editable fields, seven-pill stage changer, note-taking with timestamped logging, the queued outreach message (if any) with copy/open-in-WhatsApp/mark-sent actions, and a full **history timeline**.

4.2 Discover & Search

Tab 1 · Real web search. Browser calls the OpenAI Responses API[5] with the web_search_preview tool[6]. Returns 5-15 structured business records.

Tab 2 · Manual entry. Form for adding one lead at a time. Two buttons: add to pool or push to CRM directly.

Tab 3 · Sample batch. Generator for realistic mock leads (zone + segment + count) for testing the pipeline.

4.3 Outreach

Three sub-tabs: **Generate & queue** (per-lead personalised messages), **Outreach queue** (grouped by channel; WhatsApp tab produces real wa.me deep-links[14]), **Sent** (history with re-queue option). Marking a message sent auto-bumps the lead's stage from "New" to "Contacted". CSV exports respect LinkedIn's no-automation policy[16].

4.4 Activity

Reverse-chronological log of every workspace event — searches, lead adds, stage changes, notes added, messages queued / sent, CSV exports. Coloured icons by event type and relative timestamps.

4.5 Configure

Three sections: **Search engine · OpenAI** — API key field (stored only in browser localStorage[11]), model picker, Save / Test / Clear buttons (key managed at platform.openai.com[8]); **Score blend** — tunable weights; **Rules & metadata** — 28 reject keywords, five segment definitions, four geo zones.

5. Architecture

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■                   BROWSER (no backend)                     ■
■                                                            ■
■    index.html  ·  app.css  ·  app.js                       ■
■                          ■                                 ■
■                          ▼                                 ■
■                   localStorage (state.v5)                  ■
■                          ■                                 ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                           ■
                           ▼  External API call (user-initiated)
                    ■■■■■■■■■■■■■■■■■■■■■
                    ■  api.openai.com  ■  (Responses API +
                    ■                  ■   web_search_preview)
                    ■■■■■■■■■■■■■■■■■■■■■
```

Hosted on Vercel as a pure static site[9] — no build step, no serverless functions, no database.

Why no backend?

Privacy. The OpenAI API key never touches our server. Browser → OpenAI direct[5].

Cost. Zero infrastructure cost beyond Vercel's free static hosting[9].

Simplicity. The entire stack is HTML + CSS + JS. One ops associate can read and modify it.

Portability. State lives in localStorage[11] — easy to export, easy to reset, no database to migrate.

Why vanilla JavaScript?

No build step. Edit a line, refresh the browser, see the change.

No framework lock-in. Future maintainers can read it without learning React/Vue/Svelte.

Lightweight. Entire JS payload is ~85 KB unminified. Uses the standard Fetch API[12].

6. The Discovery Engine (OpenAI Integration)

The real-search workflow runs entirely in the browser. When the user types a query and hits search:

```
1.  Browser checks: is an OpenAI key configured?
    ■■ If no → friendly error with link to Configure view

2.  Browser POSTs to https://api.openai.com/v1/responses          [5]
    Headers: Authorization: Bearer <user's key>
             Content-Type: application/json
    Body:    { model: 'gpt-4o-mini', input: <prompt>,
               tools: [{ type: 'web_search_preview' }],            [6]
               tool_choice: 'auto' }

3.  Prompt instructs OpenAI to:
    - Search the live web for businesses matching the query
    - Find 5-15 REAL businesses (verified via web search)
    - SKIP anything resembling apartments / towers / multi-storey
    - Return strict JSON with structured business fields
    - Map each result to one of C-DISC's five segments

4.  Browser strips markdown fences, parses JSON, renders results

5.  Local G+1 keyword scan runs as a belt-and-braces second layer
```

```
6.  Each result remains in the pool until the user explicitly
    selects it and clicks 'Add to CRM'
```

Error handling.

HTTP 401: Invalid API key — friendly link to Configure.

HTTP 429: Rate limit hit — wait, retry.

Network timeout (28s): Friendly fallback message.

Malformed JSON: Graceful error with retry option.

Cost. Each search call typically costs ~$0.001 (gpt-4o-mini) to ~$0.02 (gpt-4o), per OpenAI's published pricing[7].

7. The CRM Data Model

Every CRM lead is one JavaScript object in state.crm with this shape:

```
{
  id:          "crm_lq9p3z_a8b2",
  name:        "Saraswati Poultry Farm",
  contact:     "Ravi Sharma",
  phone:       "+91 98XXXXXXXX",
  email:       "ravi@example.com",
  website:     "https://...",
  address:     "Pilibhit Road, Bareilly, UP",
  district:    "Bareilly",
  segment:     "farm_infra",      // one of 5 segments
  source:      "openai",          // openai / manual / sample
  stage:       "contacted",       // one of 7 stages
  addedAt:     1736512348000,
  lastActivity: 1736598734000,
  history: [
    { ts: ..., type: 'created',  note: 'Added from web search' },
    { ts: ..., type: 'edit',     note: 'phone: "" -> "+91 ..."' },
    { ts: ..., type: 'sent',     note: 'Sent via whatsapp' },
    { ts: ..., type: 'stage_change', note: 'New -> Contacted' },
  ],
  msgChannel:  "whatsapp",        // whatsapp / email / linkedin
  msgLang:     "hi",              // en / hi
  msgBody:     "Namaste Ravi ji, ...",
  queuedFor:   "whatsapp",
  sentAt:      1736598734000,
  notes:       ""
}
```

The seven pipeline stages:

Stage	Colour	When it changes
`new`	grey	Default when added to CRM
`contacted`	blue	Auto-set when message marked sent / WhatsApp opened
`replied`	purple	Manual — when prospect responds
`qualified`	amber	Manual — confirmed interest
`proposal`	green	Manual — quote / BoQ shared
`won`	green	Manual — closed-won
`lost`	red	Manual — closed-lost

8. The Outreach Engine — Templates Matrix

The outreach engine is built around a **30-block template matrix**: 3 channels × 2 languages × 5 segments.

```
                   Channels (3)
              ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
              ■ WhatsApp  ■  Email   ■  LinkedIn   ■
  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  ■  English   ■    ✓     ■    ✓    ■    ✓        ■
```

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■  ■■■■■■     ■    ✓      ■    ✓     ■    ✓      ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                 × 5 segments
             = 30 template blocks
```

A typical WhatsApp template (Hindi) for the industrial_shed segment:

```
Namaste {contact} ji, C-DISC ■■■■■ {name} ■■ G+1 shed ■■ ■■■
PEN panels — RCC ■■ 25-30% ■■ ■■■■, 3-4 weeks ■■■ ■■■■■■
Sample BoQ ■■■■■? YES reply ■■■■■ ■
```

Placeholders are substituted from each lead's record before display. Hindi templates use real Devanagari for emails and Hinglish (Latin script + English nouns) for WhatsApp.

The WhatsApp queue produces real wa.me deep-links per WhatsApp's click-to-chat specification[14]: https://wa.me/919XXXXXXXXX?text=<URL-encoded>. Tapping the link opens WhatsApp pre-filled for manual send. This stays **fully compliant with WhatsApp's Messaging Policy**[15] — no auto-sending, no headless bots.

9. The G+1 Enforcement — Three Defensive Layers

C-DISC's PEN Foundation is specified for ground-floor buildings only[2]. Pitching it to a multi-storey developer is a strategic failure mode the agent prevents in three layers:

Layer 1 — Local reject-keyword scan

A list of 28 reject keywords (apartment, tower, G+2-G+6, multi-storey, flats, heights, penthouse, condo, etc.) is checked against every lead name. A hit flags red, disables the checkbox, and prevents addition to CRM.

Layer 2 — Prompt instruction to OpenAI

The Responses API prompt[5] explicitly tells the model to skip multi-storey / apartment / tower / G+2+ buildings before returning results. This filters at the source.

Layer 3 — Template absence

There are no message templates for "multi-storey" or "tower" segments. Even if a rejected lead somehow leaked through the first two layers, generateMessageFor() returns null for any unknown segment.

So the agent literally cannot generate a PEN pitch to a multi-storey developer through any path.

10. State, Storage & Persistence

The entire application state lives in **one JavaScript object**, serialised to localStorage[11] under the key cdisc-north-state-v5:

```
{
  discovered: [...],     // transient pool
  crm:        [...],     // committed leads
  activity:   [...],     // event log (max 200 entries)
  weights:    { fit: 45, geo: 30, seg: 25 },
  config:     { language, channel, minScore,
                openAiKey, openAiModel },
  meta:       { totalSearches, totalAddedToCRM, lastSearch },
  theme:      'light' | 'dark'
}
```

State survives browser refresh, inspectable via DevTools → Application → Local Storage, resettable with the *Reset workspace* button. The version-bumping pattern (v1, v2, … v5) invalidates stale schemas when the data shape evolves.

No data ever leaves the user's browser except (a) the OpenAI API call [5], and (b) the optional wa.me link [14] when the user clicks to send a WhatsApp message.

11. Build History — An Honest Account

Five major iterations to reach the current shape. Each iteration corrected a real failure mode:

V1 — Landing page

Initial response treated this as a static marketing page. Useful as a starting deployment, but not an operational tool.

V2 — Operational workspace

Refactored into a single-page agent with multiple views. Had an auto-flow bug — the agent automatically dumped discovered leads straight into the CRM.

V3 — Two-stage flow

Introduced the **discovery pool / CRM separation**. Explicit user selection before leads enter the CRM. Pipeline stages introduced. Per-lead drawer with history timeline.

V4 — Real search (OSM)

Initial real-search via OpenStreetMap Nominatim, then Overpass. Two problems: Nominatim is a geocoder (empty for businesses); Vercel's serverless IPs were blocked by Overpass mirrors. Browser-direct call fixed the second.

V5 — OpenAI web search (current)

Replaced OSM entirely with OpenAI's Responses API[5] + web_search_preview[6]. Produces structured business data with contact info that Overpass could not. Added Configure view, surgical re-rendering, 28-second timeout, friendly error handling.

12. Running, Deploying, Iterating

Run locally (no install):

```
cd cdisc-app/
python3 -m http.server 8000
```

Deploy to Vercel[9]:

```
cd cdisc-app/
vercel deploy --prod --yes
```

Static site, no build step. Aliasing to a custom domain is one CLI command[10].

File map:

File	Lines	Role
`index.html`	~80	Sidebar shell + drawer + toast scaffolding
`app.css`	~1,600	Complete design system (light + dark themes, kanban, drawer)

`app.js`	~2,500	Constants, state, utilities, OpenAI integration, CRM logic, message templates, view functions
`vercel.json`	~25	Security headers + cache policies

13. Honest Limitations & Caveats

The tool is good. It is not perfect. The limitations matter:

Single-user. State is in browser localStorage[11]. No multi-user sync. Two ops associates have separate copies.

API key exposure. The OpenAI key sits in the user's browser. Anyone with devtools access can read it. Mitigation: project-scoped key[8] with low usage cap, rotate periodically.

No email delivery. The Email tab exports CSVs but doesn't send. Connecting an SMTP service (e.g. Resend / SendGrid via a Vercel function) is the natural next step.

No WhatsApp BSP. The wa.me link approach[14] is ToS-safe[15] but manual. A BSP integration would enable bulk send.

Search quality. Some queries return excellent structured data; others sparse. Varies by prompt specificity and model.

OpenAI cost. ~$0.001 to ~$0.02 per search[7]. Set OpenAI account caps.

14. Future Enhancements

• Server-side OpenAI proxy — move the API key into Vercel env vars, expose /api/search as the only public surface. Removes the key-exposure risk.

• Real email integration via a Vercel function calling Resend/SendGrid.

• WhatsApp BSP integration — replace wa.me links[14] with direct BSP send once C-DISC has a verified business profile.

• Multi-user sync — replace localStorage[11] with a Postgres/Supabase backend. Adds auth, makes the tool team-usable.

• Lead enrichment — automatic LinkedIn / website lookup, within LinkedIn's policies[16].

• Pipeline analytics — conversion-rate charts per channel, beachhead, segment.

• Mobile-first UI for the WhatsApp workflow.

• Integration with the strategy deck data — populate the KPI dashboard once C-DISC shares the [TBD] numbers.

15. Why This Matters Strategically

A strategy document tells leadership *what the right answer is*. A working tool proves that **the team building the strategy understood the operational reality well enough to make the answer executable**. That is the single biggest credibility signal a strategy team can offer.

Most North-India expansion plans for South-based construction companies have failed in the same way: a strategic recommendation lands on the leadership's desk, gets approved, and then dies in execution because the field team has no tools, no playbook, no operating layer.

The C-DISC North Engine pre-empts that failure mode by shipping the operating layer alongside the strategy. The first North-India ops hire can sit down at their laptop on day one and start working.

That is the strategic value of building the tool. The strategy deck answers *should we expand North?* The agent answers **yes, and here is how Tuesday morning looks**.

References & Footnotes

Every numbered superscript above corresponds to a reference below. Sources are organised into categories.

Company & product

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